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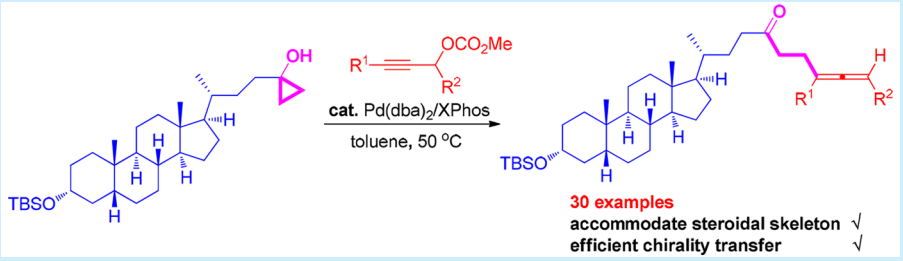
Matched Coupling of Propargylic Carbonates with Cyclopropanols

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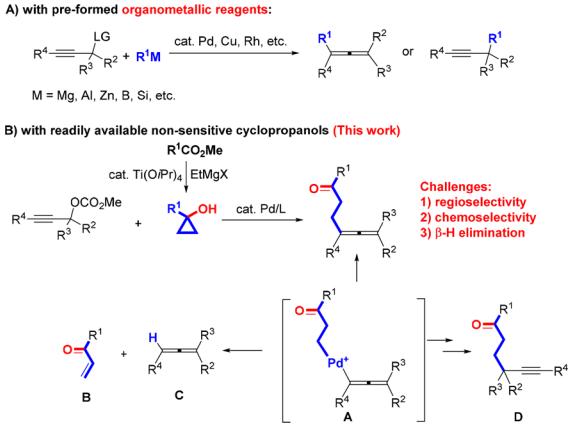
Supporting Information



ABSTRACT: The ring opening−coupling reaction of cyclopropanols with propargylic carbonates affording synthetically attractive allenyl ketones has been developed. The mechanism involves the ligand-exchange reaction of in situ formed allenyl palladium methoxide with cyclopropanols followed by carbon−carbon bond cleavage and reductive elimination. The reactions proceeded smoothly under mild reaction conditions with Pd(0)/XPhos catalysis in the absence of any external base and displayed a wide scope and application to a steroidal skeleton. The efficiency of chirality transfer and synthetic utility of the allene products have also been demonstrated.

The transition-metal-catalyzed coupling reaction of propargylic alcohol derivatives with preformed organometallic reagents such as organomagnesium, aluminum, zinc, borane, and silane reagents have been extensively studied for the synthesis of either alkynes and/or allenes[1−4] (Scheme 1A). We envisioned application to cyclopropanols, which are readily available from Ti-catalyzed reaction of carboxylates with ethylmagnesium bromide,[5,6] as an equivalent to organometallic reagents to couple with propargylic carbonates forming intermediate **A** to afford allenyl ketone products.[7] The challenges are the match of the C−C bond cleavage reaction with the oxidative addition of propargylic carbonates with palladium, the intrinsic readily occurring β-H elimination affording α,β-unsaturated enone **B** together with the corresponding reductive elimination product allene **C**, and the possible formation of regioisomeric product alkyne **D** (Scheme 1B). Here, we report our recent results on Pd-catalyzed efficient and highly selective synthesis of allenyl ketones.

Scheme 1. Different Coupling Protocols and 3,4-Allenyl Ketone Synthesis



Cyclopropanol **1a** and propargylic carbonate **2a** were taken as the model substrates to test our hypothesis. After some trial and error, toluene was the solvent of choice for this transformation. Subsequently, different types of phosphine ligands were screened. Unfortunately, $Pd(dba)_2$ and PPh_3 provided merely 13% yield of the desired allene product **3aa**, and the **B**-type of β-H elimination product enone **4′** and **C**-type of allene **3aa′** were observed as the major products. In addition, an unexpected ring-opening−protonolysis product, ethyl benzyl ketone **4**, was also formed (Table 1, entry 1).

When other ligands, such as PCy_3, *rac*-BINAP, LB-Phos, and Gorlos-Phos, were used, the substrate **2a** was recovered with various amounts of the byproducts **3aa′**, **4**, and **4′** (Table 1, entries 2−5). To our delight, when SPhos was used as ligand, the corresponding allene product was generated in 58% yield, although **3aa′** was also formed in 20% yield (Table 1, entry 6). Furthermore, changing ligand to XPhos provided **3aa** in 78%

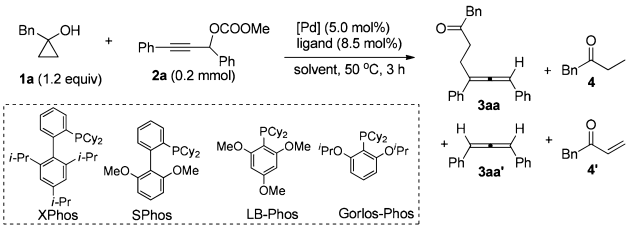
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Table 1. Optimization of Reaction Conditions[a]



entry	[Pd]	ligand	yield[b] (3aa, %)	yield[b] (3aa′/4/4′, %)	recovery[b] (2a, %)
1[c]	$Pd(dba)_2$	PPh_3	13	54/5/51	0
2[c]	$Pd(dba)_2$	PCy_3	0	5/6/<3	87
3[c]	$Pd(dba)_2$	*rac*-BINAP	0	0/4/<3	62
4[c]	$Pd(dba)_2$	LB-Phos	0	27/6/8	61
5[c]	$Pd(dba)_2$	Gorlos-Phos	0	9/8/13	67
6[c]	$Pd(dba)_2$	SPhos	58	20/3/6	0
7	$Pd(dba)_2$	XPhos	78	5/4/<3	0
8[d]	$Pd(dba)_2$	XPhos	73	10/13/4	0
9	$Pd(OAc)_2$	XPhos	73	10/15/<3	0
10	$Pd(TFA)_2$	XPhos	54	4/12/6	26
11[e]	**$Pd(dba)_2$**	**XPhos**	**84**	**6/4/8**	**0**
12[f]	$Pd(dba)_2$	XPhos	59	5/5/8	22

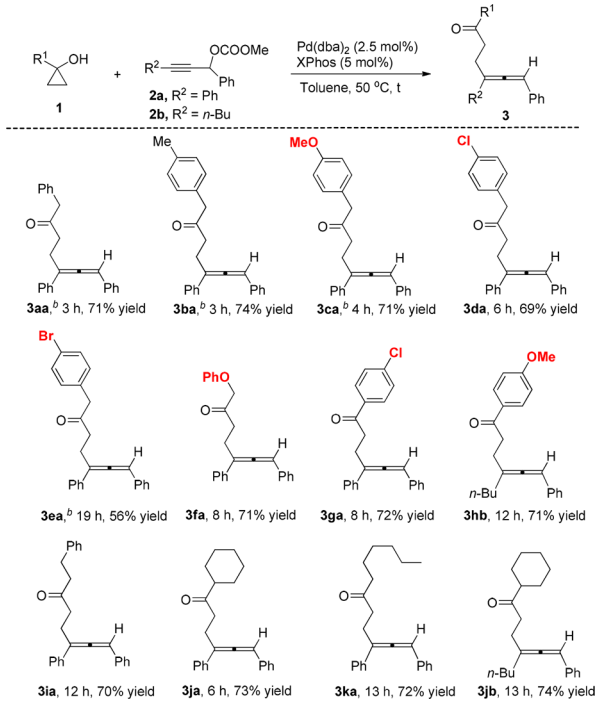
[a]Reaction conditions: **1a** (0.24 mmol), **2a** (0.20 mmol), [Pd] (5.0 mol %), ligand (8.5 mol %), in solvent (1.0 mL) at 50 °C. [b]NMR yield with CH_2Br_2 as the internal standard. [c][Pd] (5.0 mol %), ligand (10.0 mol %) was used. [d]Cs_2CO_3 (1.5 equiv) was added. [e][Pd] (2.5 mol %), XPhos (5.0 mol %) was used. [f]**1a** (0.48 mmol), **2a** (0.40 mmol), [Pd] (1.0 mol %), XPhos (2.0 mol %), in toluene (2.0 mL) reacted for 6 h.

yield, and the formation of byproducts **3aa′**, **4**, and **4′** can be suppressed to a very low level (Table 1, entry 7). Control experiments showed that Cs_2CO_3 could not improve either the yield or selectivity of the reaction (Table 1, entry 8). The results with $Pd(OAc)_2$ and $Pd(TFA)_2$ were worse (Table 1, entries 9 and 10). Finally, the loading of $Pd(dba)_2$ and XPhos may be reduced to 2.5 and 5 mol %, which even provided a slightly higher yield of **3aa** with a similar selectivity (Table 1, entries 11 and 12). It should be noted that **D**-type alkyne product was NOT formed in all cases.

With the optimized reaction conditions in hand (Table 1, entry 11), the generality of this reaction with the scope of differently substituted cyclopropanols was investigated, and the results are summarized in Scheme 2. Different substituents, such as Me and synthetically useful MeO, Cl, and Br on the benzyl group could be tolerated, providing the corresponding 3,4-allenyl ketone products **3aa**–**ea** in 56–74% yields. The X-ray structure of **3ea** confirmed the allene structure unambiguously (Figure 1).[8] Phenoxymethyl and 4-Cl- or 4-MeO-substituted aryl groups all worked smoothly to provide the final products **3fa**, **3ga**, and **3hb** in 71–72% yields. R^1 may also be different alkyl groups, such as $PhCH_2CH_2$, *n*-hexyl, or cyclohexyl, delivering the desired allene products **3ia**–**ka** and **3jb** in 70–74% yields.

Subsequently, different propargylic carbonates were also investigated with cyclopropanol **1a** (Scheme 3). R^2 may be different alkyl groups such as *n*-butyl, *n*-hexyl, and $PhCH_2CH_2$ (**3ab**–**ad**, 76–81% yields). Different functional groups such as C=C bond, chloro, ketone, ester, and cyano groups may all be incorporated into R^2 and R^3 to provide the final products **3ae**–**aj** in 55–79% yields. In addition to these secondary propargylic carbonates, tertiary propargylic carbonates may also be applied to

Scheme 2. Substrate Scope Referred to Cyclopropanols[a]



[a]Reaction conditions: **1** (1.2 mmol), **2a** or **2b** (1.0 mmol), $Pd(dba)_2$ (2.5 mol %), and XPhos (5.0 mol %) in toluene (5.0 mL) at 50 °C. [b]$Pd(dba)_2$ (5.0 mol %) and XPhos (8.5 mol %) were used.

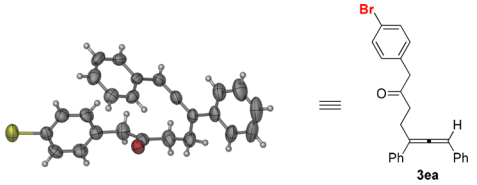


Figure 1. ORTEP representation of **3ea**.

afford the fully substituted allene products **3ak** and **3al**; however, the reaction conditions had to be reengineered: $K_3PO_4 \cdot 3H_2O$ should be applied and the reaction should be conducted at 80 °C. When R^2 = H, the corresponding disubstituted allene product **3an** was also obtained albeit in a very low yield (20%).

Since some allenes bearing steroidal skeleton have been showing pharmacological activities,[9] cyclopropanol bearing a steroid unit **1l** was applied under the standard reaction conditions. To our delight, the reaction with two representative propargylic carbonates **2a** and **2d** worked smoothly to provide the corresponding allene products **3la** and **3ld** in high yields (Scheme 4).

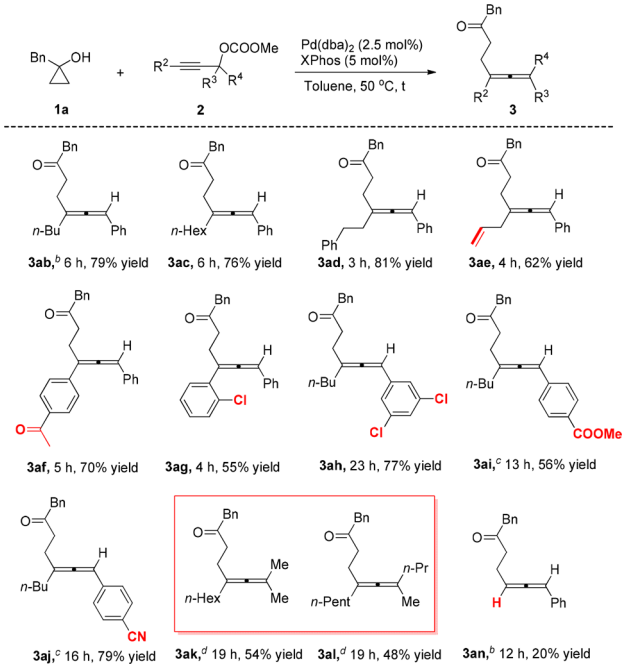
Furthermore, we studied the efficiency of central to axial chirality transfer by applying optically active chiral propargylic carbonates (*R*)-**2e**, (*R*)-**2g** and (*R*)-**2m** (Scheme 5). The corresponding chiral allenes were generated smoothly with high er. The absolute configuration of the product was determined by X-ray crystal structure of enantiopure (*R*)-**3am** (Figure 2).[10]

The potential of this reaction was further demonstrated on gram-scale synthesis of the allene product **3db** in 75% yield. This allene product **3db** may be subjected to various types of transformations, such as reduction of the ketone group with $NaBH_4$ providing the secondary alcohol **5**, 1,2-addition of the

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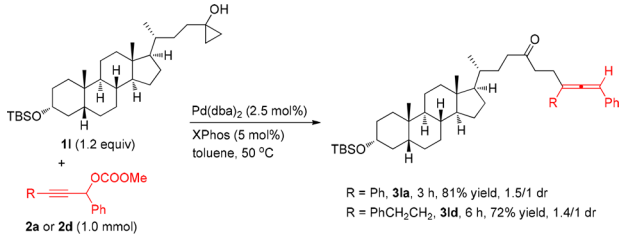
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Scheme 3. Substrate Scope Referred to Propargylic Carbonates[a]



[a]Reaction conditions: **1a** (1.2 mmol), **2** (1.0 mmol), $Pd(dba)_2$ (2.5 mol %), and XPhos (5.0 mol %) in toluene (5.0 mL) at 50 °C. [b]$Pd(dba)_2$ (5.0 mol %) and XPhos (8.5 mol %) were used. [c]**1a** (1.0 mmol) was used. [d]$Pd(dba)_2$ (2.5 mol %), XPhos (5.0 mol %), and $K_3PO_4 \cdot 3H_2O$ (1.0 equiv) in toluene (5.0 mL) at 80 °C.

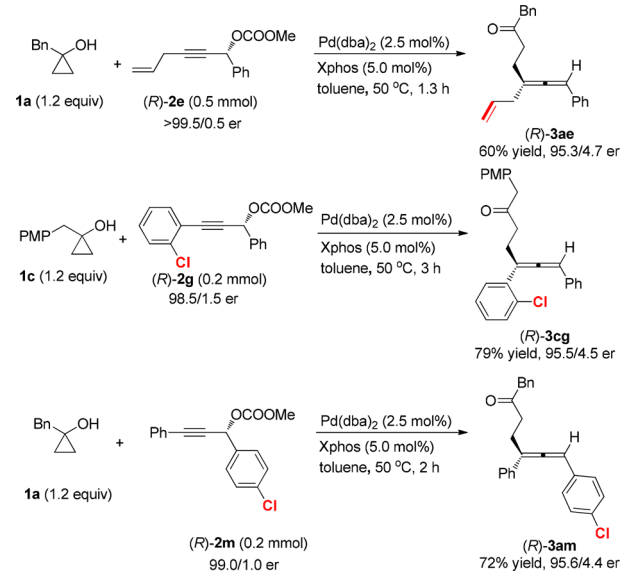
Scheme 4. Synthesis of Allenes Containing a Steroidal Skeleton



ketone group with EtMgBr or alkynyl lithium reagent generating the tertiary alcohols **6** and **7**, and reductive amination yielding allenyl amine **8**. The one-pot reduction-Au(I)-catalyzed cyclization was also conducted providing the trisubstituted tetrahydrofuran derivative **12** in a decent yield (Scheme S1 in the SI file).

To shed some light on the reaction mechanism, several control experiments were carried out (Scheme 6). First, when cyclopropanol **1a** was submitted to the standard conditions, **1a** was recovered in 84% yield together with 6% of ring opening protonolysis product **4**, and β-H elimination product **4′** was not observed, indicating that most likely the reaction did not start with **1a** (Scheme 6, eq 1). Second, when carbonate **2a** was treated with ethanol instead of cyclopropanol **1a**, both alkyne **9a** and allene **3aa′** were formed respectively, as a mixture, indicating the unique effect of the cyclopropanols on the high selectivity for the exclusive formation of allene. Although a very slow reaction was observed when *tert*-butyl alcohol was used, both **9a** and **3aa′** were detected with a similar ratio as with EtOH as judged by ^{1}H NMR analysis and isolation; thus, steric effects were not the major

Scheme 5. Chiral Allene Synthesis via Chirality Transfer



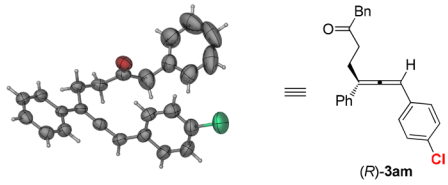
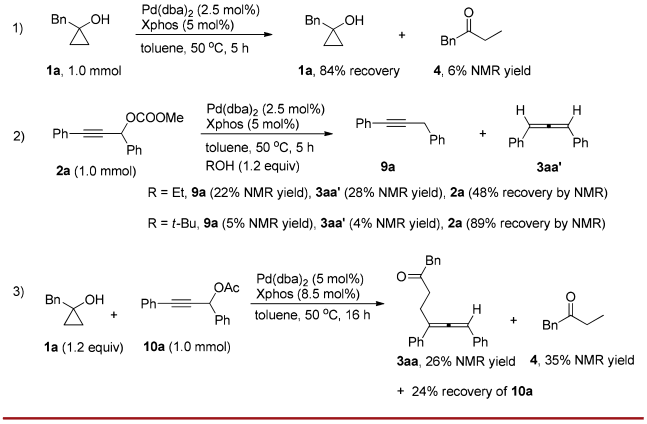


Figure 2. ORTEP representation of (*R*)-**3am**.

Scheme 6. Mechanistic Studies



reason for the high selectivity of allene over alkyne (Scheme 6, eq 2).

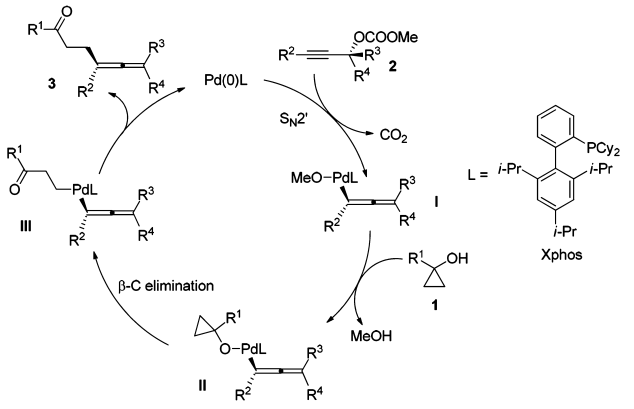
Finally, the coupling of propargylic acetate **10a** with **1a** was found to be not only quite sluggish but also to yield the related allene product **3aa** in only 26% NMR yield together with 35% NMR yield of ketone **4** and 24% of **10a**, which highlighted the importance of the carbonate for the high selectivity of this coupling reaction matched reactivity of carbonates with the ring-opening reaction of cyclopropanols (Scheme 6, eq 3).

On the basis of these results and the stereochemical outcome of the chirality transfer shown in Scheme 5, a plausible mechanism is proposed (Scheme 7): S_N2'-type *anti*-oxidative addition of Pd(0) with propargylic carbonate **2** would provide, by eliminating a molecule of CO_2, the allenyl palladium methoxide intermediate **I**, which would undergo ligand exchange with cyclopropanol **1** providing the intermediate **II**. Subsequent ring

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Scheme 7. Proposed Mechanism



opening reaction of the three-membered cycle involving a β-C elimination would yield intermediate **III**, which would undergo reductive elimination to deliver the final product **3** and regenerate the Pd(0) species to complete the catalytic cycle. We reason that the steric bulkiness of the ligand (XPhos) prevents intermediate III from β-H elimination and facilitates the subsequent reductive elimination, thus, avoiding the formation of **B**- and **C**-type of products shown in Scheme 1.

In conclusion, a palladium-catalyzed highly selective and efficient synthesis of allenyl ketones via the matched oxidative addition of propargylic carbonates with Pd(0) and the ring opening of cyclopropanols has been developed. The reaction features mild reaction conditions and a wide substrate scope without using any air sensitive metal reagents or external base, allowing the efficient introduction of the allene unit into useful organic skeleton including even the steroidal skeleton. The high efficiency of chiral transfer and synthetic utility of the 3,4-allenyl ketone products were also demonstrated.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.7b03637.

Experimental procedure, spectroscopic data (PDF)
^{1}H and ^{13}C NMR spectra of all products (PDF)

Accession Codes

CCDC 1555049 and 1579063 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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(8) Crystal data for compound **3ea**: $C_{25}H_{21}BrO$, MW = 417.33, monoclinic, space group $P2(1)/c$, final R indices [$I > 2\sigma(I)$], R_1 = 0.0295, wR_2 = 0.0738; R indices (all data), R_1 = 0.0436, wR_2 = 0.0811, a = 16.2736(6) Å, b = 14.3858(5) Å, c = 8.6941(3) Å, α = 90.00°, β = 94.7750(10)°, γ = 90.00°, V = 2028.30(12) Å^3, T = 296(2) K, Z = 4, reflections collected/unique 23191/3559 [R_{int} = 0.0267], no. of observations [$> 2\sigma(I)$] 2806, parameters: 244.

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(10) Crystal data for compound (*R*)-**3am**: $C_{25}H_{21}ClO$, MW = 372.87, orthorhombic, space group $P2(1)2(1)2(1)$, final R indices [$I > 2\sigma(I)$], R_1 = 0.0569, wR_2 = 0.1299; R indices (all data), R_1 = 0.1404, wR_2 = 0.1736, a = 7.1672(7) Å, b = 8.0143(8) Å, c = 35.791(3) Å, α = 90.00°, β = 90.00°, γ = 90.00°, V = 2055.8(3) Å^3, T = 296(2) K, Z = 4, reflections collected/unique 24220/3633 [R_{int} = 0.1128], number of observations [$> 2\sigma(I)$] 3633, parameters: 233.

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